Name: ____________________________

Date: ______________________________

Mr. Rawson • Y9 MYP Physics

1.7 – The Speed of Sound in Different Substances

Unit 1: Oscillations & Waves

Learning intentions

- a) I can explain why the speed of sound varies between gases, liquids and solids.
- b) I can compare approximate speeds of sound in air, water and steel.
- c) I can calculate time using distance = speed x time for sound waves.
- d) I can create a labelled bar chart to present researched data on sound speeds.

Theory

The **speed of sound** is the distance that a sound wave travels through a medium per unit of time. This speed is not constant; it depends heavily on the **medium** (the substance the sound travels through). In general, sound travels fastest in solids, slower in liquids, and slowest in gases. This happens because sound is a mechanical wave that relies on particles bumping into each other to pass energy along. In solids, particles are packed closely together and held by strong bonds, so vibrations transfer very quickly. In gases like air, particles are far apart, so it takes longer for the vibration to jump from one particle to the next.

Typical approximate values for the speed of sound at room temperature are: - Air: roughly 330 m/s - Water: about 1500 m/s - Steel: about 5000 m/s

To calculate how long it takes for sound to travel a certain distance, we use the equation: time $= \frac{\text{distance}}{\text{speed}}$

For example, imagine a swimmer at one end of a 25 m pool claps their hands. How long does it take for the sound to reach the other side through the water? 1. Identify the known values: distance (d) = 25 m, speed in water (v) ≈ 1500 m/s. 2. Rearrange the formula: time$(t) = d/v$. 3. Substitute the values: $t = 25/1500$. 4. Calculate: $t \approx 0.0167$ s.

When presenting data about sound speeds, a **bar chart** is often useful because it allows for easy comparison between different categories (like air vs water vs steel). A good bar chart must have a clear **title**, labelled **axes** with units, and bars of equal width. Always record the **sources** (websites or books) where you found your data to ensure accuracy.

Complete the sentence: Sound travels fastest in solids because the particles are ___________ together, allowing vibrations to pass on more quickly.

Word bank: closer · faster · slower · further · weaker

Activity

Comparing Speeds of Sound

You will research the speed of sound in three different materials and present your findings visually.

- Use a reliable science website or textbook to find the approximate speed of sound (in m/s) for:
- Air (at $20°$C)
- Water (freshwater, at $20°$C)
- Steel
- Record your findings in the table below. If you cannot find an exact value for steel, use the approximate value given in the Theory section (5000 m/s).

Material	Speed of Sound (m/s)	Source URL or Book Title
Air	____________	____________
Water	____________	____________
Steel	____________	____________

- Draw a bar chart on paper or using software to compare these speeds.
- Label the x-axis ”Material” and the y-axis ”Speed (m/s)”.
- Ensure your scale is appropriate (e.g., if the max value is 5000, your axis might go up to 6000).
- Write a clear title for your chart.

Questions

1. Explain in one sentence why sound travels faster in steel than in air, referring to particle arrangement.

2. Calculate the time it takes for sound to travel 330 m through air. Show your working.

3. A lightning strike is seen 5 s before the thunder is heard. Assuming the speed of sound in air is 330 m/s and light travels instantly, how far away was the lightning?

4. Why would a bar chart be more useful than a line graph for comparing the speed of sound in air, water, and steel?

5. If the speed of sound in water were actually 1480 m/s instead of 1500 m/s, how would this change the time calculated for the swimming pool example in the Theory section? (Show your calculation).

Extension

Research the speed of sound in diamond (≈ 12000 m/s) and compare it to steel. Explain why the speed is so much higher in diamond, considering its atomic structure and bond strength.